

Problem 1:

Let R be a ring, $S \subset R$ a multiplicative system, and M an R -module.

(a)

Formulate and prove a universal property for the natural R -module homomorphism $M \rightarrow S^{-1}M$, analogous to our universal property for the localization homomorphism $R \rightarrow S^{-1}R$ for rings.

(Universal Property):

Let N be a R -module s.t. every $s \in S$ acts invertibly (i.e. $n \mapsto sn$ is an automorphism).

Then for every R -linear map $f : M \rightarrow N$ there exists a unique $S^{-1}R$ -linear map $g : S^{-1}M \rightarrow N$ s.t. the diagram commutes

$$\begin{array}{ccc} M & \xrightarrow{\pi} & S^{-1}M \\ & \searrow f & \downarrow g \\ & & N \end{array}$$

Claim

For $S^{-1}R$ a localization of R if N is a R -module where $s \in S$ acts invertibly on N then the action

$$(r/s)n = h_s(rn) = n(r/s)$$

where h_s is the inverse map of $n \mapsto sn$ makes N a $S^{-1}R$ module.

Proof.

Well-Defined:

If $r/s = r'/s'$ then there exists a $t \in S$ s.t. $t(rs' - r's) = 0$. So for n in N

$$0 = t(rs' - r's)n = t(rs'n - r'sn)$$

Thus since t acts invertibly,

$$rs'n = r'sn$$

$$rn = sh_{s'}(r'n)$$

$$h_s(rn) = h_{s'}(r'n)$$

Thus, $h_s(rn)$ only depends on the equivalence class of r/s , so the action is well-defined.

Module:

Let $r/s, r'/s' \in S^{-1}R$ and $n, m \in N$ then

1.

$$(r/s)(n + m) = h_s(r(n + m)) = h_s(rn + rm) = h_s(rn) + h_s(rm) = (r/s)n + (r/s)m$$

2.

$$\begin{aligned} (r/s + r'/s')n &= \left(\frac{rs' + r's}{s's}\right)n = h_{s's}(rs'n + r'sn) \\ &= h_s(h_{s'}(rs'n + r'sn)) = h_s(rn) + h_{s'}r'n = (r/s)n + (r'/s')n \end{aligned}$$

3.

$$(r/s \cdot r'/s')n = (rr'/ss')n = h_{s'}(r'h_s(rn)) = h_{s'}(r'(r/s)n) = (r'/s')((r/s)n)$$

4.

$$1/1 \cdot n = h_1(n) = n$$

Therefore, N can be seen as a $S^{-1}R$ module under this action, and so any R -module which has all elements $s \in S$ act invertibly on it this implies it is a $S^{-1}R$ -module. ■

Proof.

Existence: Shown in class $S^{-1}M$ is a well-defined $S^{-1}R$ module where all $s \in S$ act invertibly.

This is true since for $m/t \in S^{-1}M$ we have that $s^{-1}s(m/t) = m/t$ and $ss^{-1}(m/t) = mt$

Let N be a R -modules s.t. every $s \in S$ acts invertibly.

Let $f : M \rightarrow N$ be a R -linear map.

Define: $g : S^{-1}M \rightarrow N$ s.t. $g(m/s) = h_s^{-1}f(m) = f(m)/s$ under the action defined above where h_s^{-1} is the inverse of the map $n \mapsto sn$.

Well-Defined: Suppose $m/s = m'/s'$ in $S^{-1}M$. Then there exists a $t \in S$ s.t.

$$t(s'm - sm') = 0$$

Now apply f

$$\begin{array}{ccc} M & \xrightarrow{\pi} & S^{-1}M \\ & \searrow f & \downarrow g \\ & & N \end{array}$$

$$t(s'f(m) - sf(m')) = 0$$

Since t acts invertible apply its inverse, and we get

$$\begin{aligned}
 s'f(m) &= sf(m') \\
 h_s(s'f(m)) &= h_s(sf(m')) \\
 s'h_s(f(m)) &= f(m') \\
 h_{s'}s'(h_s(f(m))) &= h_{s'}(f(m')) \\
 h_s(f(m)) &= h_{s'}(f(m')) \\
 g(m/s) &= g(m'/s')
 \end{aligned}$$

Homomorphism:

Let $m/s, m'/s' \in S^{-1}M$

Since $s'sn = s'(sn) \implies h_{s's} = h_s \circ h_{s'}$

$$\begin{aligned}
 g\left(\frac{m}{s} + \frac{m'}{s'}\right) &= g\left(\frac{s'm + sm'}{ss'}\right) = h_{ss'}(s'f(m) + sf(m')) \\
 &= h_{s'}(s'h_s(f(m))) + h_{s'}(h_s(f(m'))) = h_s(f(m)) + h_{s'}(f(m')) = g(m/s) + g(m'/s')
 \end{aligned}$$

Let $m/s \in S^{-1}M$ and $r/t \in S^{-1}R$.

$$\begin{aligned}
 g(r/t \cdot m/s) &= g(rm/ts) = g(rm/st) = h_{st}(f(rm)) = h_t(h_s(rf(m))) \\
 &= h_t(rh_s(f(m))) = (r/t)(g(m/s))
 \end{aligned}$$

Thus, g is $S^{-1}R$ -linear.

Diagram:

Let $m \in M$ then $g(\pi(m)) = g(m/1) = h_1(f(m)) = f(m)$. Thus, $g \circ \pi = f$.

Uniqueness:

Let $h : S^{-1}M \rightarrow N$ be $S^{-1}R$ -linear with $h \circ \pi = f$. For $m/s \in S^{-1}M$,

$$h(m/s) = (1/s)h(\pi(m)) = (1/s)f(m) = h_sf(m) = g(m/s)$$

Thus, g is unique. ■

(b)

Prove that if M is an R -module such that every element $u \in S$ acts invertibly on M , then there is an isomorphism $M \cong S^{-1}M$ of R -modules.

Proof.

Assume M is an R module s.t. for all $s \in S$ s acts invertibly on M . Let h_s be the inverse of $m \mapsto sm$.

Define $\pi : M \rightarrow S^{-1}M$ s.t. $\varphi(m) = m/1$

By the claim you can view M as a $S^{-1}R$ module with the action $(r/s)m = h_s(rm)$.

Using the universal property this implies that there is a unique map $S^{-1}R$ -linear map s.t. $\psi : S^{-1}M \rightarrow M$ s.t. $\psi \circ \pi = \text{id}$.

Again using the universal property with $S^{-1}M$ we get there exists a unique map $\theta : S^{-1}M \rightarrow S^{-1}M$ s.t. $\theta \circ \pi = \pi$

Since id also works this implies that $\theta = \text{id}$

Also, $(\pi \circ \varphi) \circ \pi = \pi$ also works this implies $\pi \circ \varphi = \text{id}$.

Thus, π and φ are $S^{-1}R$ -linear maps and are inverses.

Thus, $S^{-1}M \cong M$ as $S^{-1}R$ modules.

But the same maps work for R modules since $\varphi(rm/s) = r\varphi(m/s)$ and $\pi(rm) = rm/1 = (r)(m/1)$.

Thus, $S^{-1}M \cong M$ as R modules. ■

(c)

Let $f : M \rightarrow N$ be an R -module homomorphism. Show that there is a natural $S^{-1}R$ -module homomorphism $S^{-1}f : S^{-1}M \rightarrow S^{-1}N$ induced by f , given by $\frac{m}{s} \rightarrow \frac{f(m)}{s}$ for $s \in S$ and $m \in M$.

Proof.

Well-Defined:

Let $m/s = m'/s'$ this implies there exists a $t \in S$ s.t. $t(s'm + sm') = 0$

$0 = f(t(s'm + sm')) = t(s'f(m) + sf(m'))$ since f is R -linear.

Therefore, this implies $f(m')/s' = f(m)/s$ thus well-defined.

Homomorphism:

Let $m/s, m'/s' \in S^{-1}M$

$$S^{-1}f(m/s + m'/s') = S^{-1}f\left(\frac{ms' + m's}{ss'}\right) = f(ms' + m's)/s's = s'f(m)/s's + sf(m')/s's = f(m)/s + f(m')/s = S^{-1}f(m/s) + S^{-1}f(m'/s').$$

Let $r/t \in S^{-1}R$

$$S^{-1}f(r/t \cdot m/s) = S^{-1}f(rm/ts) = rf(m)/ts = (r/t) \cdot f(m)/s = (r/t) \cdot S^{-1}f(m/s)$$

Thus, $S^{-1}f$ is a $S^{-1}R$ -module homomorphism. ■

Problem 2:**(b) Localization is Exact**

Let $S \subset R$ be a multiplicative system. Given an exact sequence $L \xrightarrow{f} M \xrightarrow{g} N$, prove the sequence of $S^{-1}R$ -module homomorphisms

$$S^{-1}L \xrightarrow{S^{-1}f} S^{-1}M \xrightarrow{S^{-1}g} S^{-1}N$$

is also exact.

Proof.

Assume $L \xrightarrow{f} M \xrightarrow{g} N$ is an exact sequence of modules.

Let $x \in \text{Im}(S^{-1}f)$ thus $f(l)/s = x$ for some $l \in L$.

So, since the sequence is exact this implies that $f(l) \in \ker(g)$. Thus, $S^{-1}g(f(l)/s) = 0/s = 0 \implies x \in \ker(S^{-1}g)$.

Let $x \in \ker(S^{-1}g)$ then $x = m/s$ for $m \in M$ and $s \in S$, so $S^{-1}g(m/s) = g(m)/s = 0$. This implies there exists a $t \in S$ s.t. $tg(m) = 0$. Since g is a R -module homomorphism this implies that $tg(m) = g(tm)$. Thus, $tm \in \ker g = \text{Im}(f)$. So there is an $l \in L$ s.t. $f(l) = tm$. So,

$$\frac{m}{s} = \frac{tm}{ts} = \frac{f(l)}{ts} = S^{-1}f\left(\frac{l}{ts}\right)$$

This shows $m/s \in \text{Im}(S^{-1}f)$.

Therefore, $\text{Im}(S^{-1}f) = \ker(S^{-1}g)$, so this is an exact sequence. ■

(c) Exactness is a local property

For any sequence $L \xrightarrow{f} M \xrightarrow{g} N$ of R -modules homomorphisms, prove that the following are equivalent:

- i. The sequence $L \xrightarrow{f} M \xrightarrow{g} N$ is exact.
- ii. The sequence $L_{\mathfrak{p}} \xrightarrow{f_{\mathfrak{p}}} M_{\mathfrak{p}} \xrightarrow{g_{\mathfrak{p}}} N_{\mathfrak{p}}$ is exact for all $\mathfrak{p} \in \text{Spec}(R)$.
- iii. The sequence $L_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} M_{\mathfrak{m}} \xrightarrow{g_{\mathfrak{m}}} N_{\mathfrak{m}}$ is exact for all maximal ideals $\mathfrak{m} \in \text{mSpec}(R)$.

Proof.

(i.) $\implies$ (ii.) from part (b). (ii.) $\implies$ (iii.) since maximal ideals are a subset of $\text{Spec}(R)$.
 (iii.) $\implies$ (i.):

Assume (iii.)

Let $x \in \text{Im}(f)$ then there exists an $l \in L$ s.t. $f(l) = x$. Let $A = \{r \in R \mid rg(x) = 0\}$.

This is an ideal since if $a, b \in A$ then $(a + b)g(x) = ag(x) + bg(x) = 0$ and for $a \in A$ and $r \in R$ we have $rag(x) = r \cdot 0 = 0$, so A is an ideal.

Assume A is a proper ideal then A is in some maximal ideal $\mathfrak{m}$. By assumption, we know that $g_m(f_m(l/1)) = 0$. This implies that $g(f(l))/1 = 0/1$, so there exists some $a \in R \setminus \mathfrak{m}$ s.t. $ag(f(l)) = 0$ but this is a contradiction since A is a subset of $\mathfrak{m}$.

Thus, A is not proper and $A = R$, thus $g(x) = 0 \implies x \in \ker g$.

Let $x \in \ker g$. Let $A = \{r \in R \mid rx \in \text{Im}(f)\}$. This is an ideal for similar reasons as above. Assume that this is a proper ideal.

Thus, there exists a maximal ideal $\mathfrak{m}$ that contains A .

Since $x/1 \in \ker(g_m) = \text{Im}f_m$ by assumption, there exists a $l/s \in L_m$ s.t. $f_m(l/s) = f(l)/s = x/1$. This implies there exists a $a \in R \setminus \mathfrak{m}$ s.t.

$a(f(l) - ns) = 0 \implies f(al) = asx$, so $asn \in \text{Im}(f)$. Thus, $as \in A$, but since $a \in R \setminus \mathfrak{m}$ and $s \in R \setminus \mathfrak{m}$ this implies that $as \in R \setminus \mathfrak{m}$ since $\mathfrak{m}$ is prime, but this is a contradiction.

Thus, $A = R$, so $1 * x = x \in \text{Im}(f)$.

Thus, $\ker(g) = \text{Im}(f)$, so the sequence is exact. ■

Problem 4:

Let (R, m) be a local domain (i.e. a local ring which is also a domain) which is not a field. Let K be the fraction field of R , which is naturally an R -algebra via the inclusion $R \subset K$. Prove that K is not a finite R -algebra.

Proof.

Assume K is a finite R -algebra. Proven in class this implies that K is integral.

This implies that for all $1/r \in K$ s.t. $r \neq 0 \in R$ then there exists $r_0, \dots, r_{n-1} \in R$ s.t. $(1/r)^n + r_{n-1}(1/r)^{n-1} + \dots + r_0 = 0$.

Multiplying by r^n gives $1 + r_{n-1}r + \dots + r_0r^n = 0$. This implies that $1 \in (r) \implies (r) = R$. Thus, r is a unit, but this implies that R is a field since r was arbitrary.

This is a contradiction thus K is not a finite R -algebra. ■

Problem 5:

Let R be a domain with fraction field K . The normalization $\widetilde{R}$ of R is the integral closure of R in K , i.e. $\widetilde{R} = \{y \in K \mid y \text{ is integral over } R\}$, and we say that R is normal if $R = \widetilde{R}$. Prove that any UFD is normal.

Proof.

Let $x = a/b \in \widetilde{R}$ with $a, b \in R$ s.t. $b \neq 0$, and $\gcd(a, b) = 1$.

Then there exists $r_{n-1}, \dots, r_0 \in R$ s.t. $x^n + r_{n-1}x^{n-1} + \dots + r_0 = 0$.

Multiply by b^n

$$a^n + r_{n-1}ba^{n-1} + \dots + r_0b^n = 0$$

$$a^n = -b(r_{n-1}a^{n-1} + \dots + r_0b^{n-1})$$

This implies that $b \mid a^n$ in R . Since R is a UFD factor a, b into irreducibles. Then if $p \mid b$ for some irreducible p . Then $p \mid a^n$, so $p \mid a$ since p is irreducible. But this is a contradiction since $\gcd(a, b) = 1$, so b is a unit and $x \in R$.

Therefore, $\widetilde{R} = R$. ■

Problem 6:**(a)**

Show that $\mathbb{C}[x, y]/(y^3 - x^5)$ is a domain which is not normal, and compute its normalization.

Proof.

$$\mathbb{C}[x, y]/(y^3 - x^5) \cong \mathbb{C}[t^3, t^5].$$

Consider $x^3 - t^3 = 0$ for $x = t$. This implies $t \in \overline{\mathbb{C}[t^3, t^5]}$. Thus, $\mathbb{C}[t] \subset \overline{\mathbb{C}[t^3, t^5]}$, but since $\mathbb{C}[t]$ is a UFD this implies it is normal thus $\overline{\mathbb{C}[t^3, t^5]} = \mathbb{C}[t]$. So $\mathbb{C}[t^3, t^5]$ is not normal, with normalization $\mathbb{C}[t]$. ■

(b)

Compute the normalization of $\mathbb{Z}[\sqrt{-3}]$

Claim

If α is integral over R then $R[\alpha]$ is finite.

Proof.

Assume α is integral over R . This implies there exists a $r_{n-1}, \dots, r_0 \in R$ s.t.

$$\alpha^n + r_{n-1}\alpha^{n-1} + \dots + r_0 = 0$$

This implies that for α^k for $k \geq n$ are a R -linear combination of $1, \dots, \alpha^{n-1}$, so $\{1, \dots, \alpha^{n-1}\}$ generate $R[\alpha]$ as an R -module. Thus, $R[\alpha]$ is finite over R . ■

Proof.

Consider ω where ω is the cube root of unity.

Since $\omega^2 + \omega + 1 = 0$ this implies that ω is integral over $\mathbb{Z}$, so $\mathbb{Z}[\omega]$ is integral over $\mathbb{Z}$ by the proof above.

Since $\mathbb{Z} \subset \mathbb{Z}[\sqrt{-3}] \subset \mathbb{Z}[\omega]$ this implies $\mathbb{Z}[\omega]$ is integral over $\mathbb{Z}[\sqrt{-3}]$ since the same monic equation in $\mathbb{Z}$ will work in $\mathbb{Z}[\sqrt{-3}]$.

Thus, $\mathbb{Z}[\omega] \subseteq \overline{\mathbb{Z}[\sqrt{-3}]}$

Since $\mathbb{Z}[\omega]$ is a UFD by problem 5, this implies that $\mathbb{Z}[\omega]$ is normal thus integrally closed. Therefore, $\mathbb{Z}[\omega] = \overline{\mathbb{Z}[\omega]}$.

Let $a \in \overline{\mathbb{Z}[\sqrt{-3}]}$. This implies that there exists $r_{n-1}, \dots, r_0 \in \mathbb{Z}[\sqrt{-3}]$ s.t. $a^n + r_{n-1}a^{n-1} +$

$\cdots + r_0 = 0$. Since $\mathbb{Z}[\sqrt{-3}] \subset \mathbb{Z}[\omega] \implies r_i \in \mathbb{Z}[\omega]$.

Thus, $a \in \widetilde{\mathbb{Z}[\omega]} = \mathbb{Z}[\omega]$.

Therefore, $\mathbb{Z}[\omega] = \mathbb{Z}[\sqrt{-3}]$. ■

Problem 7:**(a) Normalization commutes with localization**

Let R be a domain and $S \subset R$ a multiplicative system. Via the inclusion $R \hookrightarrow \widetilde{R}$, we can also regard S as a multiplicative system in the normalization $\widetilde{R}$. Prove that there is an isomorphism of rings

$$S^{-1}\widetilde{R} \cong \widetilde{S^{-1}R},$$

where $\widetilde{S^{-1}R}$ is the normalization of $S^{-1}R$.

Proof.

Let $K = \text{Frac}(S^{-1}R) \cong \text{Frac}(R)$ since R is a domain, and we are given $S^{-1}R$ is a domain. Consider $\varphi : S^{-1}\widetilde{R} \hookrightarrow K$, so $S^{-1}\widetilde{R}$ is isomorphic to a subring of K and $\widetilde{S^{-1}R} \subset K$.

Let A be the isomorphic subring.

Since $\widetilde{R}$ is integral over R this implies that for $x = r/s \in S^{-1}\widetilde{R}$. Thus $r \in \widetilde{R}$, so there exists a monic polynomial s.t. $r^n + r_{n-1}r^{n-1} + \cdots + r_0 = 0$ for $r_i \in R$.

Thus, in $S^{-1}\widetilde{R}$, $(r^n/1) + r_{n-1}(r^{n-1}/1) + \cdots + r_0 = 0 \implies x^n + r_{n-1}/s \cdot x^{n-1} + \cdots + r_0/s^n = 0$

Thus, $S^{-1}\widetilde{R}$ is integral over $S^{-1}R$. This implies since A is isomorphic that A is integral over $S^{-1}R$ thus $A \subset \widetilde{S^{-1}R}$.

Let $x \in \widetilde{S^{-1}R}$. This means that x is integral over $S^{-1}R$, then

$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ for $a_i \in S^{-1}R$.

Let $a_i = c_i/s_i$. Let $d = \prod_{i=0}^{n-1} s_i \in S$ be a common denominator then multiply by d^n . We get

$(dx)^n + c_{n-1}(dx)^{n-1} + \cdots + c_0 = 0$ for $c_i \in R$.

Thus, dx is integral over R , so $dx \in \widetilde{R} \implies sx/1 \cdot 1/s = x \in A$. Thus, $A = \widetilde{S^{-1}R}$.

Therefore, $S^{-1}\widetilde{R} \cong A = \widetilde{S^{-1}R}$ ■

(b) Normality is a local property

Let R be a domain. Prove that the following are equivalent:

- i. R is normal
- ii. $R_{\mathfrak{p}}$ is normal for all $\mathfrak{p} \in \text{Spec}(R)$.
- iii. $R_{\mathfrak{m}}$ is normal all $\mathfrak{m} \in \text{mSpec}(R)$

Proof.

(i) $\implies$ (ii) by (a) since if R is normal then $S^{-1}R = S^{-1}\widetilde{R} \cong \widetilde{S^{-1}R}$.

(ii) $\implies$ (iii) since maximal ideals are a subset of prime ideals.

(iii) $\implies$ (i):

Assume (iii).

Let $x \in \widetilde{R}$ then there exists $r_0, \dots, r_{n-1} \in R$ s.t. $x^n + r_{n-1}x^{n-1} + \dots + r_0 = 0$.

Let $I = \{a \in R \mid ax \in R\}$ where ax is multiplication in R 's fraction field.

Assume I is proper, so there exists a maximal ideal $\mathfrak{m}$ that contains I .

Consider $R_{\mathfrak{m}}$ we can view the relation x has in $R_{\mathfrak{m}}$ thus

$x^n + r_{n-1}/1 \cdot x^{n-1} + \dots + r_0/1 = 0$ thus $x \in R_{\mathfrak{m}}$ since by assumption $R_{\mathfrak{m}}$ is normal.

Let $x = r/s$ for $r \in R$ and $s \in R \setminus \mathfrak{m}$.

This implies that $(sx)^n + r_{n-1}s(sx)^{n-1} + \dots + s^n r_0 = 0$ $sx \in \widetilde{R}$ and $sx \in R$.

Thus, $s \in I$ but this is a contradiction since $s \in R \setminus \mathfrak{m}$ and I is contained in $\mathfrak{m}$. Thus, $I = R$ and $1x = x \in R$.

Thus, $\widetilde{R} = R$. ■